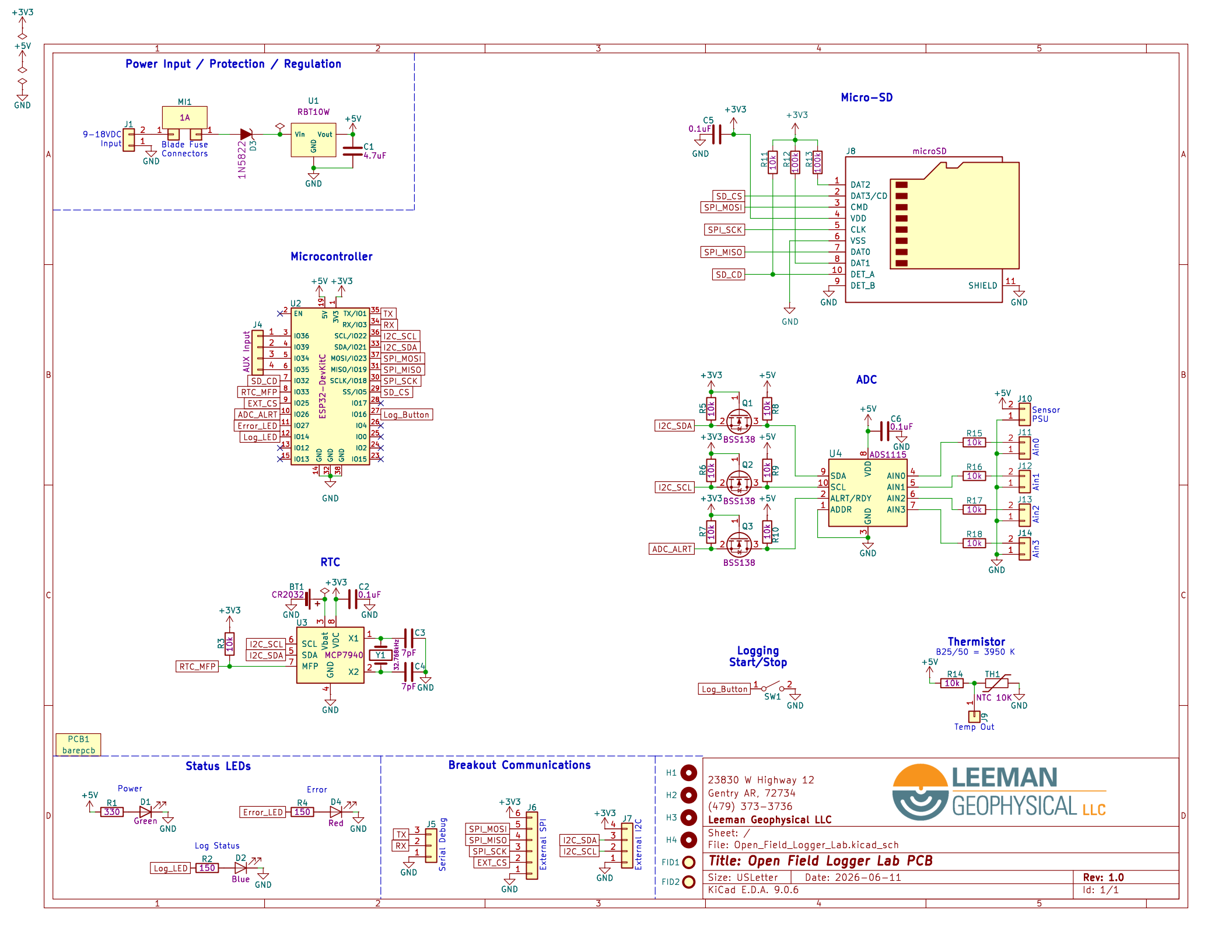
Power Input / Protection / Regulation
MI1
1A
J1
9–18VDC Input
Blade Fuse Connectors
1N5822
D3
U1
RBT10W
Vin
Vout
GND
+5V
C1
4.7uF
Micro-SD
C5
0.1uF
R11
10k
R12
100k
R13
100k
J8
microSD
SD_CS
SPI_MOSI
SPI_SCK
SPI_MISO
SD_CD
DAT2
DAT3/CD
CMD
VDD
CLK
VSS
DAT0
DAT1
DET_A
DET_B
SHIELD
Microcontroller
U2
ESP32-DevKitC
J4
AUX Input
TX
RX
I2C_SCL
I2C_SDA
SPI_MOSI
SPI_MISO
SPI_SCK
SD_CS
Log_Button
RTC_MFP
EXT_CS
ADC_ALRT
Error_LED
Log_LED
ADC
Q1
Q2
Q3
BSS138
U4
ADS1115
SDA
SCL
ALRT/RDY
ADDR
AIN0
AIN1
AIN2
AIN3
C6
0.1uF
J10
Sensor PSU
J11
Ain0
J12
Ain1
J13
Ain2
J14
Ain3
R15
R16
R17
R18
RTC
BT1
CR2032
C2
0.1uF
U3
MCP7940
SCL
SDA
MFP
Vbat
VDC
X1
X2
Y1
32.768kHz
C3
7pF
C4
7pF
R3
10k
Logging Start/Stop
Log_Button
SW1
Thermistor
B25/50 = 3950 K
R14
10k
TH1
NTC 10K
J9
Temp Out
PCB1
barepcb
Status LEDs
Power
R1
330
D1
Green
Error
R4
150
D4
Red
Log Status
R2
150
D2
Blue
Breakout Communications
J5
Serial Debug
J6
External SPI
J7
External I2C
H1
H2
H3
H4
FID1
FID2
23830 W Highway 12
Gentry AR, 72734
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Leeman Geophysical LLC
LEEMAN GEOPHYSICAL LLC
Sheet: /
File: Open_Field_Logger_Lab.kicad_sch
Title: Open Field Logger Lab PCB
Size: USLetter
Date: 2026-06-11
Rev: 1.0
KiCad E.D.A. 9.0.6
Id: 1/1